

Unbreakable Cryptosystems

Computational security

A cryptosystem is considered to be computationally secure if the best algorithm that can be used for breaking it requires at least N steps/operations, where N is a very, very, ridiculously large positive integer. We can't prove that any cryptosystem is computationally secure because we can never assert that we know what the "best" algorithm is; for example some evil attacker may have cooked up an algorithm that the security industry doesn't know about.

We can however use the N value as a benchmark for brute force attacks such as an exhaustive key search.

Provable Security

Cryptosystems whose security is based on solving a known problem are provably secure; any evil attacker who can efficiently break the cryptosystem must be able to efficiently solve the known problem.

For example, the RSA cryptosystem is based on the integer factorization problem. If an evil attacker can efficiently break RSA, then as a consequence they must be able to also efficiently factor an integer.

Note that the proof of security for a provably secure cryptosystem is *relative to the known problem* and not absolute or an absolute proof of security.

Unconditional security

The holy grail of security. A cryptosystem that is unconditionally secure cannot be broken, even with infinite computing power. But the cost of eternal security comes at a price; the length of any key used must be at least as long as the plaintext message, and keys can only be used once.

This creates key management issues that tend to make these types of cryptosystems impractical for general use.

One-time Pad

An n -bit One-time Pad cryptosystem (M, C, K, E, D) has $M = C = K = (\mathbb{Z}_2)^n$, the set of all binary strings of length n .

$x \in (\mathbb{Z}_2)^n$, $y \in (\mathbb{Z}_2)^n$, $k \in (\mathbb{Z}_2)^n$; in other words, the plaintext, ciphertext, and keys are all from the set of all plaintext binary strings of length n . This implies that the length of the key (n -bits) is equal to the length of the plaintext (n -bits) and the length of the ciphertext (n -bits).

$e_k(x) = (x_1 x_2 \dots x_n) \oplus (k_1 k_2 \dots k_n)$; the encryption rule for all keys is the bit-wise XOR of the plaintext x , and the key k .

$d_k(y) = (y_1 y_2 \dots y_n) \oplus (k_1 k_2 \dots k_n)$; the decryption rule for all keys is the bit-wise XOR of the ciphertext y , and the key k .

Keys are randomly generated and used one only once (hence the name “One-time”).

Example:

Encrypt the hex string “BB8” using the binary key “0101 1111 0001”.

We have that $x = 1011\ 1011\ 1000 \in (\mathbb{Z}_2)^{12}$ and that $k = 0101\ 1111\ 0001 \in (\mathbb{Z}_2)^{12}$.

Encrypting we obtain:

$$\begin{aligned} y &= e_k(x) \\ &= (1011\ 1011\ 1000) \oplus (0101\ 1111\ 0001); \\ &= 1110\ 0100\ 1001 \end{aligned}$$

Or in hex, the ciphertext is “E49”.

Proof of perfect secrecy of the One-time pad

Recall that a cryptosystem has perfect secrecy if $P(x | y) = P(x)$ for all $x \in M$ and $y \in C$.

For the n -bit One-time Pad cryptosystem we have that keys are randomly generated and used one only once. This means that each time we encrypt, the probability of using a given n -bit key is $P(k_i) = \frac{1}{2^n}$ for $1 \leq i \leq n$. The probability of getting a given ciphertext y_0 is:

$$\begin{aligned} P(y = y_0) &= P(k_1) \cdot P(y = y_0 | k_1) + P(k_2) \cdot P(y = y_0 | k_2) + \dots + P(k_n) \cdot P(y = y_0 | k_n) \\ &= \frac{1}{2^n} \cdot P(y = y_0 | k_1) + \frac{1}{2^n} \cdot P(y = y_0 | k_2) + \dots + \frac{1}{2^n} \cdot P(y = y_0 | k_n) \\ &= \frac{1}{2^n} \left[\sum_{i=1}^n P(y = y_0 | k_i) \right] \end{aligned}$$

We also have that $P(y = y_0 | k_i) = P(x_i)$, where x_i is the plaintext such that $y_0 = e_{k_i}(x_i)$ (in other words, encrypting x_i with key k_i gives the ciphertext y_0). In general we don't know the individual probabilities $P(x_i)$ (because that would mean knowing which plaintext binary strings are more frequent than others), but we do know that they all sum to 1. Therefore:

$$P(y = y_0) = \frac{1}{2^n} \left[\sum_{i=1}^n P(y = y_0 | k_i) \right] = \frac{1}{2^n} \left[\sum_{i=1}^n P(x_i) \right] = \frac{1}{2^n} [1]$$

The probability of getting a given ciphertext y_0 given a plaintext x_0 is $P(y = y_0 | x = x_0) = P(k_i) = \frac{1}{2^n}$.

This is because there only exists one unique key, k_i , that encrypts x_0 as y_0 .

Using Baye's theorem we have that:

$$\begin{aligned} P(x = x_0 | y = y_0) &= \frac{P(x = x_0) \cdot P(y = y_0 | x = x_0)}{P(y = y_0)} \\ &= \frac{P(x = x_0) \cdot \frac{1}{2^n}}{\frac{1}{2^n} [1]} \\ &= P(x = x_0) \end{aligned}$$

Which proves the condition for perfect secrecy. Therefore, the n -bit One-time Pad is unconditionally secure against a ciphertext only attack.

Exercises:

1. One-time Pad encryption and decryption:

- Encrypt the hex string "AB89" using the binary key "1100 1001 0111 1110".
- Decrypt the hex string "465" using the binary key "1111 1100 1000".
- Decrypt the lower case letter string "toefxllmd" using the lower case letter key "mathrules" by converting them both to $\mathbb{Z}_{26}$ and subtracting the key.

2. The following cryptosystem consists of the following plaintext, ciphertext, and key sets:

$x \in M = \{a, b\}$, the set of all plaintext strings

$y \in C = \{0, 1\}$, the set of all ciphertext strings

$k \in K = \{k_1, k_2\}$, the set of all encryption keys

The probability distributions for x and k are:

$$P(x = a) = 0.9, P(x = b) = 0.1 \text{ and } P(k_1) = \frac{1}{2}, P(k_2) = \frac{1}{2}$$

The encryption matrix is:

	a	b
k_1	1	0
k_2	0	1

Prove that this cryptosystem is unconditionally secure against a ciphertext only attack.

3. The following cryptosystem consists of the following plaintext, ciphertext, and key sets:

$x \in M = \{q, r\}$, the set of all plaintext strings

$y \in C = \{0, 1\}$, the set of all ciphertext strings

$k \in K = \{k_1, k_2\}$, the set of all encryption keys

The probability distributions for x and k are:

$$P(x = q) = 0.3, P(x = r) = 0.7 \text{ and } P(k_1) = \frac{1}{5}, P(k_2) = \frac{4}{5}$$

The encryption matrix is:

	q	r
k_1	1	0
k_2	0	1

Prove that this cryptosystem is vulnerable to a ciphertext only attack.

4. A single element shift cipher consists of the following plaintext, ciphertext, and key sets:

$x \in M = \mathbb{Z}_{26}$, the set of all plaintext strings

$y \in C = \mathbb{Z}_{26}$, the set of all ciphertext strings

$k \in K = \mathbb{Z}_{26}$, the set of all encryption keys

$e_k(x) = (x + k) \bmod 26$; the encryption rule for all keys.

$d_k(x) = (x - k) \bmod 26$; the decryption rule for all keys.

The probability distribution for k is: $P(k) = \frac{1}{26}$ for each $k \in \mathbb{Z}_{26}$.

Prove that this cryptosystem is unconditionally secure against a ciphertext only attack.

Answers:

1.

- a) 62FE
- b) BAD
- c) holygrail